Birdy

A physical bird voice, a corpus with no audio in it, and a label on every claim

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Abstract. Birdy is an instrument in three parts. It sings with a physical model of the bird syrinx: two equations driven by air-sac pressure and labial tension, fitted to field recordings by gesture inversion. It remembers 340 079 measured syllables, held as numbers and never as audio. It shows the result as a three-dimensional map with one point per measured syllable. Its claim about meaning is narrow and stated on the face of it: call type to documented behavioural function, with a confidence label on every phrase it speaks. Playback experiments are the only proof of meaning the field accepts. Birdy has run none of its own, so everything past the literature is labelled as inference or as invention. This note reports the instrument and its evidence. The learned syllable embedding lifts species placement by 13.3 points. The parallel learn is bit-identical to the serial one and about twelve times faster. A session-quality gate separates pathological recordings at AUC 0.953. Three measurements run the other way. The number the pipeline optimises carries almost no information about audible distortion ($r \approx 0.1$). The embedding organises by microphone before it organises by bird (+0.488 against +0.196). And more data buys recorder invariance more cheaply than any loss term tried so far. Section 9 lists every pre-registered gate and its result in one table. Refutations stay in the record. That is the method.

Keywords. birdsong · syrinx model · gesture inversion · contrastive embedding · bioacoustics · confidence labels · negative results

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1 What Birdy is, and what it claims

> A great tit does not improvise. It owns a small set of calls, and one of them is a sentence: three notes that mean scan for danger, then one note that means come here. In that order, other tits do both things. Played backwards, they do nothing. So birdsong is structure, and a machine can learn structure. Meaning is the harder half. The field accepts one proof of it, the playback experiment — play the sound to a bird and watch what the bird does. Birdy is built on both halves. It learns the structure hard, and it labels the meaning honestly.

The field's own position has three parts[1]. Machine learning today can **characterise** a repertoire: find the units and their structure. It can **functionally decode**: link a signal to a context and a behaviour. It **cannot** translate into propositional human language, because that needs playback experiments, and those exist for a handful of systems. An ethological asymmetry sharpens the point. Bird *song* has rich syntax and weak referential meaning. Bird *calls* carry meaning and little syntax (docs/03 §3.1). The defensible product is therefore **functional translation**: call type to documented behavioural function, plus species, individual identity and urgency, each with a confidence label. Word-for-word translation is marketing.

The vocabularies behind the language set are real, and each one is somebody's fieldwork. The Japanese great tit composes ABC–D: ABC means scan for danger, D means approach, and the reversed order elicits nothing[2]. The Siberian jay's alarm calls encode predator *behaviour* rather than predator species[3]. The chickadee's chick-a-dee call has four note types, a strict A→B→C→D order, and a D-count that scales with threat[4]. The nightingale delivers 180 to 200 song types in rule-governed order[5]. The Bengalese finch sings a finite-state syntax[6]. That literature-anchored set is what Birdy calls a documented function. The five labels below cover everything outside it.

The honesty rule: five labels, and every claim carries one. The rule (docs/08 §8.5) is a product requirement, and it is the reason the rest of this note can be read at face value. This section states it once, and the sections after it report measurements. Every phrase the app speaks carries its confidence: *experimental* for playback-verified, *documented* for the literature, *inferred* for plausible and unestablished, and *invented* for a designed mapping, always named as designed. Anything the system flags as new gets the fifth label, **"unlabelled: real acoustics, no documented function"**. Detection reports that a sound is new. The label claims exactly that much and stops there.

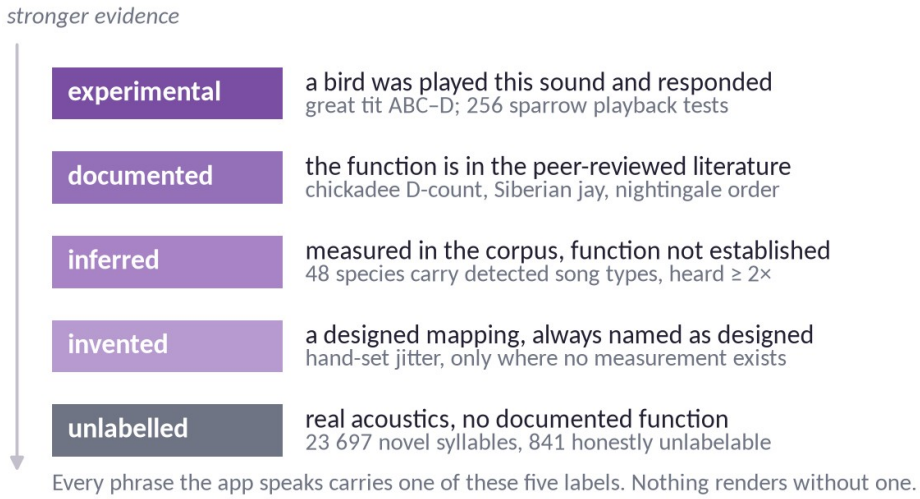


Figure 1 — The claim ladder. Five levels of evidence, and the label the interface prints for each. The counts are the current corpus. Source: docs/08 §8.5, docs/12, docs/16 §16.5.

Why a physical synthesiser. Two reasons carry the design, one scientific and one legal. Scientifically, a procedural model of the syrinx is a falsifiable theory of the bird, because its parameters — air-sac pressure, labial tension, tract geometry — are physiological claims that can be wrong. The model class has passed the strongest realism tests the field has. Synthetic song from it drives song premotor neurons indistinguishably from real song[13], and wild territorial sparrows respond to it as they do to a real rival[20]. Legally, it is the only clean voice available, for the reason section 3 sets out. Every sound the app makes is synthesised from fitted parameters, and every point in its map is a measurement.

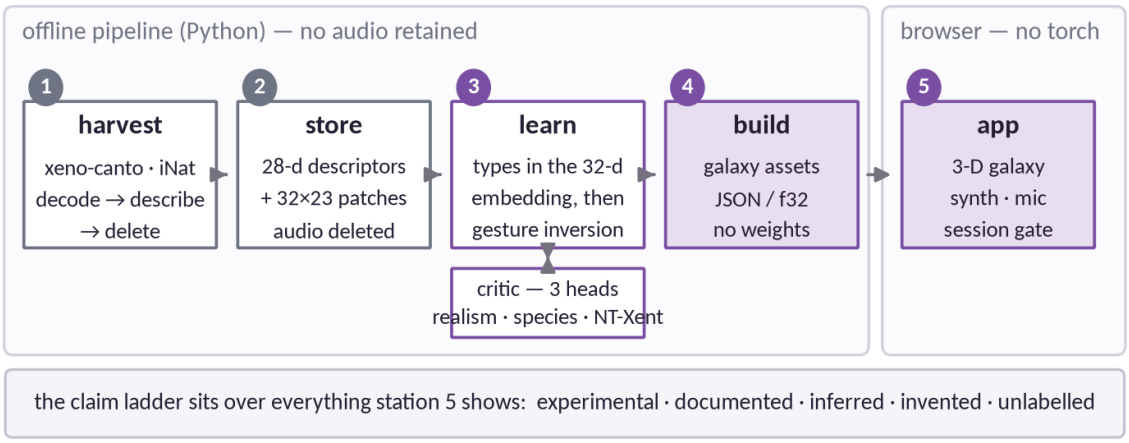


Figure 2 — The instrument, end to end. Stations 1 to 4 run offline in Python and are the subject of sections 3 and 4. Station 5 is the browser: static assets, no torch, no Python, described in section 5. Sources: docs/06 §6.1, docs/08 §8.3.

Table 1 is the legend to that figure. It names every technique, the place it runs, and the reason it is there. Rows marked parked are implemented and parity-tested, and they ship off by default. Section 2 gives their measured cost.

component	technique	where it runs	why it is there
voice source	Mindlin–Laje flapping oscillator, eq. 1	AudioWorklet synth + Python fit	physiological, bird-validated, real-time (docs/08 §8.1)
time scale	per-syllable γ, $f_0 = g \cdot \Phi$, eq. 2	fit + synth	Φ(0.1, 1) = 0.17061 is measured, not fitted (docs/08 §8.1)
tract	f_0-tracking resonance + stopped-tube comb, eq. 3	synth	pure tone is a filter effect; the comb ships in 10 of 22 species (docs/08 §8.1a)
roughness	140 Hz noise on β	synth	tension noise drives forebrain neurons, and an intermediate amount scores best (docs/08 §8.1a)
fricatives	turbulence routed to fitted formants	synth, per type	a chip note is shaped noise, not a tone with hiss (docs/16 §16.7)
respiration	air-sac oscillator, eq. 4	parked, off by default	the hand-set default is a wash; the per-species fit won 445 types (docs/15 §1)
two voices	sides coupled via tracheal pressure	parked, off by default	hand-set default regresses 0.343 → 0.294 (docs/15 §5)
descriptor	28-d: 13 MFCC means + 13 stds + centroid + flatness	harvest, fit, browser	the one representation Python and the browser can compute identically (docs/08 §8.2)
spectral fingerprint	32×23 log-mel patch, 736 numbers	store, critic	not invertible to audio, so the licensing property holds (docs/09 §9.3)
critic	3-head CNN: realism, species, contrastive	pipeline	it scores and types, and it never generates (docs/09 §9.2b)
contrastive loss	NT-Xent, τ = 0.2, eq. 5	critic training	same syllable close, different syllables apart (docs/09 §9.2b)

component	technique	where it runs	why it is there
map space	the critic's 32-d embedding	build, galaxy	+13.3 points of species placement over the hand-weighted vector (docs/09 §9.8)
browser bridge	random Fourier features + ridge, eq. 6	build fits it, browser applies it	no torch in the browser; gated at cosine 0.8 (docs/11 §11.3)
fit objective	28-d descriptor cosine, eq. 7	inversion, ship gates	cluster-safe and audibly blind — section 8 (docs/08 §8.2, docs/19)
session gate	off-manifold 5-NN distance	browser live path	AUC 0.953, computable from the 28-d side alone (docs/20 §20.3.2)
grammar	first-order + Kneser–Ney backoff	build, app sequences	unseen tail +1.90 / +2.32 nats, seen side bit-identical (docs/18)
song baseline	bag-of-types	build	the bar any learned song encoder has to beat (docs/18 P2)
novel-species hint	Perch 2.0 top-3, ONNX, offline	pipeline only	metadata, never typing ground truth (docs/13)

Table 1 — Where the system uses what. Every technique, the place it runs, and the note that defines it.

2 The voice: a syrinx, and the audit of it

> Birds do not sing with a larynx. At the bottom of the windpipe sits the syrinx: soft membranes that vibrate in the airstream like a reed. The bird has two controls. Air-sac pressure sets how hard it blows, which is roughly loudness. Labial tension sets how tight the membranes are, which is roughly pitch. Two equations for that reed, driven by those two controls, are realistic enough to fool the auditory neurons of a listening bird. Birdy's voice is those equations plus the physics of the tube above them.

2.1 The source

The source is the normal-form flapping oscillator of the Mindlin lineage[7,8,9], integrated as two first-order equations (docs/08 §8.1):

$$\dot{x} = y, \qquad \dot{y} = -\alpha\, g^2 - \beta\, g^2 x - g^2 x^3 - g\, x^2 y + g^2 x^2 - g\, xy \tag{1}$$

Here α is air-sac pressure and β is labial tension, which sets the fundamental frequency. g is a pure time-scale: rescaling time by g leaves the equations unchanged. Each syllable gets its own g, chosen to place its median pitch in the well-behaved region of the model:

$$f_0 = g \cdot \Phi(\alpha, \beta), \qquad \Phi(0.1,\ 1) = 0.17061 \tag{2}$$

Φ was measured once on a calibration sweep, so the value above is a measured constant of the model[17]. The bifurcation that starts the oscillation decides the timbre. At high tension a Hopf bifurcation gives tonal, whistle-like notes. At low tension a saddle-node on a limit cycle gives pulse-like, harmonically dense ones, and the spectral content index is a

function of tension alone[14]. The tonal-to-rich axis is therefore the model's own tension scale, set at the source. The integrator is deliberately plain: forward Euler at 16× oversampling at 48 kHz. RK4 agrees to within 2 % of F0, so Euler is what ships.

2.2 The tube above it

The vocal tract does acoustic work. Pure-tone birdsong is a filter effect, in which the tract suppresses the harmonics the source produces, and heliox experiments prove it[23,24]. Songbirds also retune the oropharyngeal cavity actively, tracking the fundamental from 1.5 to 5 kHz inside a single syllable[26]. So one resonance in Birdy follows the instantaneous f_0. A tracking resonance on its own would make every spectrum pitch-invariant, and two species at the same pitch would become the same instrument. The trachea supplies the fixed part. It is a tube closed at the syrinx, and it resonates as a stopped quarter-wave pipe at the odd harmonics of one body-fixed frequency:

$$f_n = (2n-1)\,f_1, \quad n = 1\ldots4, \qquad f_1 = \tfrac{c}{4L} \tag{3}$$

The measured agreement is close. Eastern towhee resonances sit at about 2.0 and 5.5 kHz against 1.88 and 5.67 kHz predicted for its 45 mm trachea[25], the cardinal shows the 1:3:5:7 pattern[26], and the white-throated sparrow sits at about 2.2 and 6.6 kHz[27]. Birdy measures f_1 per species from the harvest itself, where pitch-dependent structure averages away and only body-fixed resonances survive. The comb ships where the resynthesis scores better with it, which was 10 of 22 species on the original corpus (docs/08 §8.1a).

2.3 Texture, which is also physics

Birdy injects turbulence at the labia and scales it by air flow, so a silent bird stays silent. Roughness perturbs β, the tension, at the source. The literature is specific here. Selective forebrain neurons required noise on tension as well as a matched fundamental and matched filters, and an intermediate amount scored best[12]. The modulation sits at 140 Hz, inside the 80 to 250 Hz band that syringeal muscle can follow[33,32]. Because the perturbation acts through the oscillator, f_0 and amplitude wobble arrive correlated, as they do in real song[36]. All noise comes from a seeded xorshift32 stream, so the pipeline and the browser generate the same turbulence.

The fricative route, which is the one extension that ships. A real chip note is a broadband burst shaped by the tract, closer to a whispered vowel than to a tone with hiss on it. The evidence came from two species already at the noise ceiling: Algerian nuthatch types scored cosine 0.16 to 0.60 with the noise parameter at 0.60 to 0.85. The fit was asking for a missing *shape*, and more noise was the wrong axis. The route sends turbulence around the f_0-tracking band into the syllable's own fitted formant set, driven by the same pressure gesture as the tonal source. The bench median moved 0.584 to 0.656, end to end 13 wins and 0 losses. It ships per type where the resynthesis beats the tonal fit, and every other syllable renders exactly as before (docs/16 §16.7).

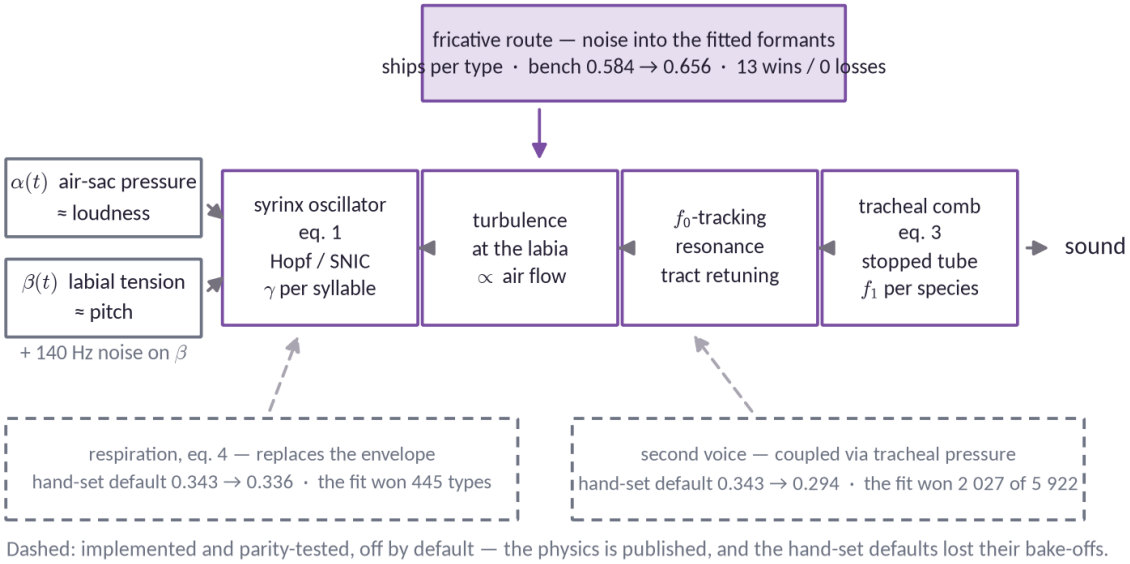


Figure 3 — The synthesis signal chain. Gesture contours drive the oscillator (eq. 1), then texture and the two tract stages shape the result. Sources: docs/08 §8.1 and §8.1a, docs/15 §1 and §5, docs/16 §16.7.

2.4 Two extensions, and what they cost

Two structural extensions are implemented, parity-tested, and off by default. The respiratory oscillator[21,22] replaces the hand-designed amplitude envelope with two-state air-sac dynamics (docs/15 §1):

$$\tau_x\,\dot{x} = -(1+x^2)\,x - p + F_0\,f(t), \qquad \tau_p\,\dot{p} = -(1+x^2)\,x - (1+\alpha(p))p + F_0\,f(t) \tag{4}$$

A Heaviside conductance asymmetry between inflow and outflow lets attack shape and cross-syllable memory emerge from the dynamics. The hand-set default lost its pre-registered bake-off: median cosine 0.343 to 0.336, with the losses concentrated in sub-50 ms notes whose onsets the physically correct 20 to 40 ms attack eats. The second extension couples the syrinx's two sides through the shared tracheal pressure[11,28,29], and it is parked for the same reason, 0.343 to 0.294 with the hand-set default, although the two worst-fitting low-pitched species improved. The per-species fit pass is the version that wins: respiration took 445 of 12 680 scored types and the second voice took 2 027 of 5 922 screened, with the ship gate holding everywhere (docs/16 §16.4). The rule this taught, twice, now applies to every extension. Hand-set physics loses bake-offs, however correct the physics is.

2.5 The audit: the model is right, and the fit is what fails

docs/21 audits the synthesiser against the canonical literature, term by term, and the verdict is worth stating plainly because it redirects the entire roadmap. The oscillator is the published normal form[8,10], with *x* sign-flipped against the papers' convention and verified numerically before the mapping was fixed. It is validated against RK4, against the reference calibration table to 0.1 % on F0 and 2 % on peak amplitude, and against a two-sided parity harness that stands at 18 of 18 fixtures. The two-voice kernel's measured strong-

coupling boundary falls exactly where its paper says the weak-coupling approximation dies[11]. The oscillator survives every measurement the project has made.

Those measurements implicate three other places. The **representation** of a syllable — one tension contour, one envelope, one static tract — cannot hold what a real note carries. The **gesture sources** are designed by hand, and section 2.4 prices that. And the **objective** is unproven as a fidelity ranking, which is section 8. The most legible symptom is a parameterisation choice. The γ rule parks every syllable's median pitch at $\beta \approx 1$, the tonal end of the model, and the output loses the spectrally rich tail.

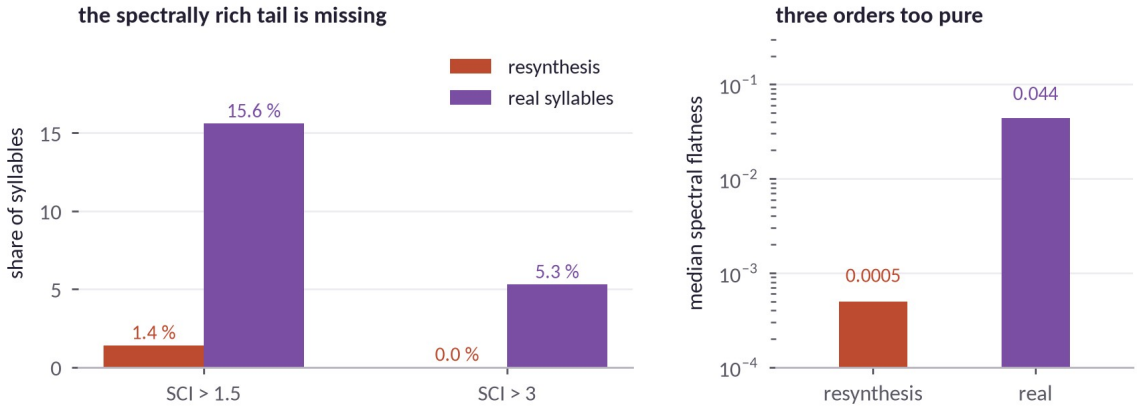


Figure 4 — The tonal-regime pin. Left: the share of syllables above two spectral-content thresholds, resynthesis against real. Right: median spectral flatness, log scale. The model contains the pulse-like regime that produces the missing tail, and the fit does not reach it. Source: docs/08 §8.1a as read by docs/21 §21.2.

A second symptom is blunter. The production learn's reachability report logged the tension clamp biting in **6 617 of 12 724 types**. Over half the fitted corpus has contour excursions that hit the tension floor or ceiling while audible, so their pitch rides the clamp instead of the dynamics. The fit is asking the equations for frequencies outside the range the calibration table measures. The fix belongs on the fit side, in per-syllable γ and wider sweeps.

Table 2 is the useful form of this. What a listener calls noise or machine decomposes into six causes, each pinned to a number the project already owns. Two of the rows are worth reading together, because noise and machine are opposite failures of one cause family. The representation has one noise axis and one envelope. A fricative collapses to unshaped hiss, which is too random. A tonal note collapses to a metronomic sine, which is not random enough. The ear files both as artificial, because real syringes live between the regimes, and the model contains that middle[14,15] while the fit stays at one end of it.

what a listener hears	the measured cause
hiss where a chip note should be	Fricative fits parked at the noise ceiling and still failing — oropendola types at noise 0.85 scoring 0.12 to 0.77. The turbulence was shaped only by an f_0-tracking band, a filter tuned to a pitch a broadband burst does not have (docs/16 §16.7).
flute purity, a whistle too clean to be an animal	The tonal-regime pin of figure 4. Real labial motion at low tension is a skewed, pulse-like waveform with rich harmonics; the regime that produces it was unreachable until gammaScale and the 0.002 calibration table (docs/08 §8.1a).

what a listener hears	the measured cause
a sine with an organ envelope	Authored one-tone calls carried noise 0, rough 0, timbre pure, while the same bird's learned types measure noise 0.05 to 0.25 and rough up to 0.65. The re-fit shipped a measured texture to 14 of 15, median gain +0.148 (docs/16 §16.4).
metronomic, the same note every time	No physiological micro-variation. The oscillator is deterministic and the envelope fixed, and variation exists only per corpus-level variant. Real crystallised song has about 1 % rendition-to-rendition frequency variation, and Birdy's roughness implements that mechanism as one static fitted depth per type.
no loudness life, onsets arrive pre-shaped	The tract is static per syllable, with formant coefficients computed once and nothing coupling pressure to filter. Real birds articulate per note, and beak gape acts as a high shelf on 4 to 7.5 kHz. The multiplicative tract drive that would fix it exists only on the default-off respiration path (docs/15 §1).
machine-fitted types sound worse than their cosine says	The objective is the suspect. The cosine does not rank audible fit, so a type can climb the proxy into a spectrally wrong corner. Median resynthesis cosine is 0.648 across the current learn, and most sub-0.3 types are a different, unfixed gap (docs/19 §19.2, docs/16 §16.3).

Table 2 — The noise-and-machine taxonomy. Six symptoms and six measured causes, all of them outside the oscillator. Source: docs/21 §21.2.

3 The corpus: decode, describe, delete

> The harvest downloads a recording, decodes it, measures every syllable in it, and deletes the file. What survives is two sets of numbers per syllable. This is the design constraint the whole project is built on, and it is worth stating once, here, in full. Xeno-canto recordings carry per-file Creative Commons licences, largely non-commercial. A corpus of measurements carries none of them, so a corpus of measurements is the one this project can build on, ship and share. One exception is owner-sanctioned: a local cache holds staged audio for the current training phase, on the machine that fits the model, and it is deleted when the phase ends.

The first measurement is a 28-dimensional descriptor: 13 MFCC means, 13 MFCC standard deviations, mean $\log_2$ spectral centroid, and mean spectral flatness. It is computed under an exact frame-level contract, 48 kHz mono with 1024/512 Hann frames, which the browser re-implements sample for sample (docs/08 §8.2). The second is a fixed 32×23 log-mel patch: 736 numbers, log magnitude only, no phase, resampled onto the syllable's own duration. The patch keeps the spectral fingerprint the critic learns from, at a resolution no decoder can return to recognisable audio (docs/09 §9.3).

The stores, because every number in this note names one. Every measurement here names the corpus it came from. Table 3 lists the stores in order, and figure 5 shows what each ingest bought.

store or phase	syllables	species	what it added
out/harvest, original	39 976	22	the reference corpus of docs/08 §8.1a
out/harvest-gan, first	9 041	25	the learned-typing bake-off store (docs/09 §9.7)
out/harvest-gan at 40 k	39 993	25, 13 after a fixed trim bug	the +13.3-point replication (docs/09 §9.8)
+ xccoverbl ingest	39 874	121	the first multi-source fetch (docs/14 §14.1)
+ iNatSounds 2024	226 264	3 811	6 453 recordings in one pass, zero failures (docs/16 §16.1)
+ P0 depth harvest	340 079	3 811	+139 440 syllables, 7 992 → 10 184 recordings (HANDOFF 2026-07-31)
current learn	340 079 embedded	2 421 fitted, 13 859 types	median resynthesis cosine 0.648, 23 minutes

Table 3 — The measurement stores. Each row is a fixed corpus. The shaded rows are the current one.

Breadth came first, and depth had to be bought separately. The iNatSounds ingest multiplied the species count by 31 and left depth per species where it was. Depth is what nearly every open problem in this note needs. The P0 depth harvest was the correction: 100 species chosen for documented syntax cases, repertoire depth and existing playback anchors, up to 30 recordings each, run in four batches of 25 with every species exiting clean. It added 139 440 syllables and took the target 100 to a median of 29 recordings per species, with 71 of them above the 20-recording threshold. The tenth percentile is 5, which is the archive's own ceiling — xeno-canto holds no more for those birds. The store grew 28.65 MB per species, all of it measurements.

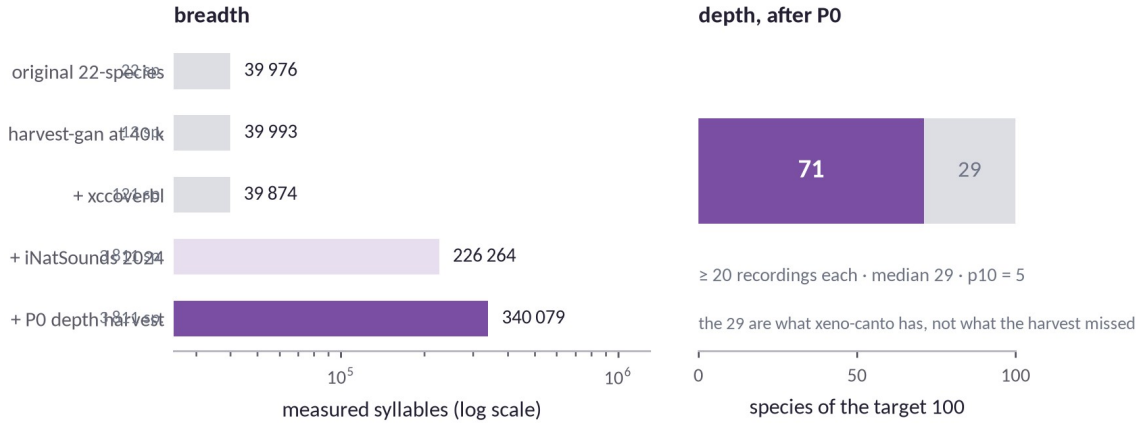


Figure 5 — Breadth, then depth. Left: measured syllables per store, log scale, with the species count at each row. Right: recordings per species on the P0 target 100. Sources: docs/08 §8.1a, docs/09 §9.7 and §9.8, docs/14 §14.1, docs/16 §16.1, HANDOFF 2026-07-31.

What the learn does with it. Syllable identity was once decided by seven hand-weighted numbers with guessed weights. It is now KMeans clustering in the critic's 32-d embedding, with the cluster count chosen by silhouette. Each type is then fitted by gesture inversion. That is a coordinate descent over about 16 axes of the synthesiser's parameter vector, in

three passes. The descriptor cosine scores each candidate against the reference observation:

$$\text{score} = \cos\left(z_{\text{syn}},\ z_{\text{ref}}\right) = \frac{z_{\text{syn}} \cdot z_{\text{ref}}}{\|z_{\text{syn}}\|\,\|z_{\text{ref}}\|} \tag{7}$$

z is the standardised 28-d descriptor in the harvest population frame. Section 8 examines this number in detail. It remains the optimiser because its failure is bounded and cluster-safe. The same pass measures rhythm. 26 897 per-transition gaps come out of it, and the app plays every gap from that table.

4 The critic, the two spaces, and the recorder in the way

> One learned component sits in the pipeline. It scores and it types, and it never generates audio. The critic is a small convolutional network over the log-mel patch. It decides which syllables are the same syllable, it flags the ones it has never heard, and its internal representation is the coordinate system of the map. It also organises that map by microphone before it organises it by bird. That is the largest unsolved problem in this note, and section 4.4 measures it.

4.1 A critic that scores, never sings

The network is four convolutional layers with spectral normalisation, about 1 to 3 million parameters, trained with three heads on a shared trunk (docs/09 §9.2b). A **realism** logit separates real patches from renders of the current fit. A **species** classifier reads the harvest labels. A **contrastive** head trains with an NT-Xent loss, so two augmented views of one syllable embed close and different syllables embed apart:

$$\ell_i = -\log \frac{\exp\left(\text{sim}(z_i, z_i^{+})/\tau\right)}{\sum_{k \neq i} \exp\left(\text{sim}(z_i, z_k)/\tau\right)}, \qquad \tau = 0.2 \tag{5}$$

Augmentations are time stretches, gain changes and added noise. Pitch shifts stay out, because pitch is type-diagnostic. Whole species are held out of training, so realism and novelty are measured on birds the network has never seen. The current critic trained on 2 686 species with 895 held out entirely. It scores realism AUC 1.000 on held-out species and species accuracy 0.730 over 2 686 classes against chance of 0.0004. Its novelty ROC is 0.846, flagging 32.2 % of unheard species' syllables at a chosen 5 % false-positive rate (docs/16 §16.2). The 32-d embedding is the deliverable. Section 8 records how the realism logit failed as a fit objective.

4.2 What the embedding bought

Typing syllables in the critic's embedding instead of the hand-weighted vector moved species placement from 26.6 % to 32.9 % on the 9 k store against chance of 4.0 %. On the 40 k store it moved placement from 38.4 % to 51.7 %, a gain of 13.3 points, against chance of 7.7 %. The guardrail cosine held at 0.807 to 0.796 and F0 error improved from 3.3 % to

2.2 %. Every tracked metric held or improved, on two independent stores (docs/09 §9.7 and §9.8). The same retrain epochs lifted novelty detection through 0.596, 0.786 and 0.846, with the true-positive rate at 5 % false positives going from 7.4 % to 32.2 %. The three ROC numbers are reported per epoch, so the whole trajectory stays visible.

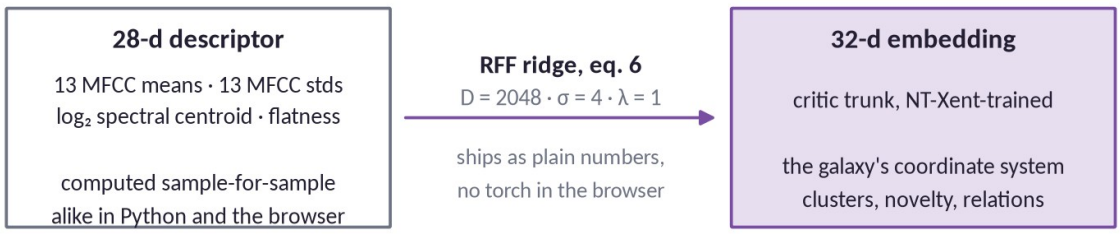
The learn that consumes it is fast and exactly reproducible. Per-species fits run in a fork pool after one GPU embedding pass, verified bit-identical to the serial fit, and the 3 800-species learn went from about three hours to about fifteen minutes on sixteen workers. It checkpoints every 100 species. The current learn on the deepened store fitted 2 421 species into 13 859 types in 23 minutes, and every fitted species carries a recovered common name (docs/16 §16.3).

4.3 Two feature spaces and a bridge

Two feature spaces coexist because of one constraint. The hand-written 28-d descriptor is what the browser can compute live. The critic's 32-d embedding is what the map is clustered in. The browser runs without torch, so a bridge maps the first space into the second: random Fourier features approximating an RBF kernel, then a closed-form ridge regression, fitted on descriptor-embedding pairs from the store with the knobs chosen on held-out recordings (docs/11 §11.3):

$$\varphi(x) = \sqrt{\tfrac{2}{D}}\cos(xR + b), \quad R \sim \mathcal{N}(0, \sigma^{-2} I), \qquad \hat{z} = \varphi(x)\,W + c \tag{6}$$

The bridge ships as plain numbers, D = 2048 features with σ = 4 and λ = 1 on the reference fit, and it ships only when it earns its place. It goes to the browser when the median mapped-versus-true cosine on held-out recordings reaches 0.8. Below that the browser works from the 28-d descriptor alone. The debug build passes at 0.962. A linear ridge came first. It reached 0.7997 once the embedding grew finer, just under the gate, so the random-Fourier version replaced it.



Ship gate: median held-out cosine ≥ 0.8. The debug build passes at 0.962; the linear ridge it replaced broke at 0.7997.

Knot-level agreement 0.61 raw · 0.91 tie-adjusted · 0.78 at species level — decomposed in figure 9.

Figure 6 — Two feature spaces and the bridge. The browser computes only the 28-d descriptor, the map lives in the critic's 32-d embedding, and the ridge of eq. 6 maps one into the other behind a ship gate. Sources: docs/08 §8.2, docs/11 §11.3, docs/20 §20.1.

4.4 The recorder in the way

The geometry battery scores how much a grouping organises the embedding, as the same-label rate among a point's nearest neighbours minus a permutation null:

$$\mathrm{lift}(G) = A(G) - \mathbb{E}_{\pi}[A(\pi G)] \tag{8}$$

On the production embedding the strongest relation is *which recordist made the recording*: kNN lift +0.488, against species at +0.196 (docs/14 §14.4). Every consumer of the embedding inherits that. The typing, the novelty detector, the map and the relations are all built on a space whose dominant axis is a microphone. The literature is unambiguous that invariance is a training-time problem[45,46,47], so Birdy attacked it there, three times, on pre-registered gates (docs/17). All three attempts on the loss side missed their gate, and each one named a reason.

Augmentation is free, and it leaves the confound where it was. Channel-EQ simulation and Freq-MixStyle on log-mel patches are perfectly species-safe, and they move recordist lift by −0.7 % relative. **The adversarial head works and costs exactly what it buys.** A recordist-adversarial head does move the target, and the trade is almost perfectly linear in log weight. At w = 0.05 it takes 33 % off recordist lift and 14.7 points off species accuracy. At w = 0.3 the embedding is destroyed, with species accuracy at 0.199 and realism AUC at 0.862. Every swept weight that reaches the required 30 % puts species accuracy outside the gate. Recorder invariance costs species information at roughly one to one (docs/17 §17.3).

The species-safe version protects species and leaves the confound standing. Masking the adversarial softmax to recordings of the patch's own species fences species information out by construction, and it does exactly that: species lift stays flat to +10 % at every weight. But it buys only −6 % against a −30 % gate. The reason is the useful part. The recordist signal is carried *between* species, because each species' recordings come from different people. That is precisely the place a species-factored head is built to ignore (docs/17 §17.5).

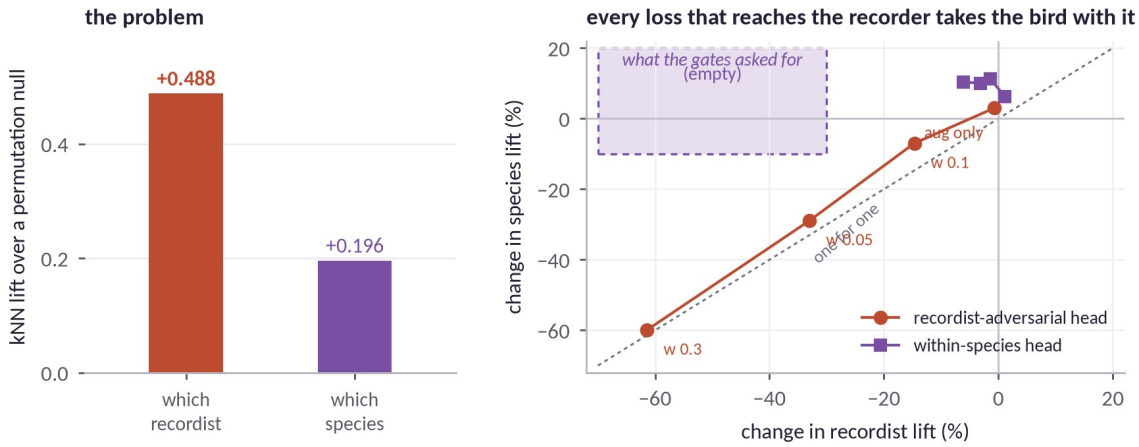


Figure 7 — The confound, and why the loss road is closed. Left: what the embedding actually organises by. Right: every swept point from both adversarial heads, as relative change in recordist lift against relative change in species lift. The shaded region is

what the pre-registered gates asked for, and no arm reaches it. 37-species debug store, debug-scale. Sources: docs/14 §14.4, docs/17 §17.3 and §17.5.

Data helps, and the honest re-test shrank the help. The AVEX direction[47] held: merging two recording domains per species beat every loss term tried. The first test used 12 species with a 3-species holdout, and the pre-registered bigger re-test used 63 both-domain species with 16 held out. The re-test shrank every delta. The naive merge's +14.4 points became +3.6. The cross-domain contrastive mechanism's +13.4-point boost became +1.7 against a +10 gate, and the AUC drop that had counted against it reversed. What survives at both scales is the ordering, and the ordering is what matters: more domains per species helps, the mechanism adds little on top, and data buys invariance more cheaply than any loss term tried (docs/17 §17.4 and §17.6). The same lesson arrives twice in this note, from two directions. Small samples flatter whichever arm you hoped for.

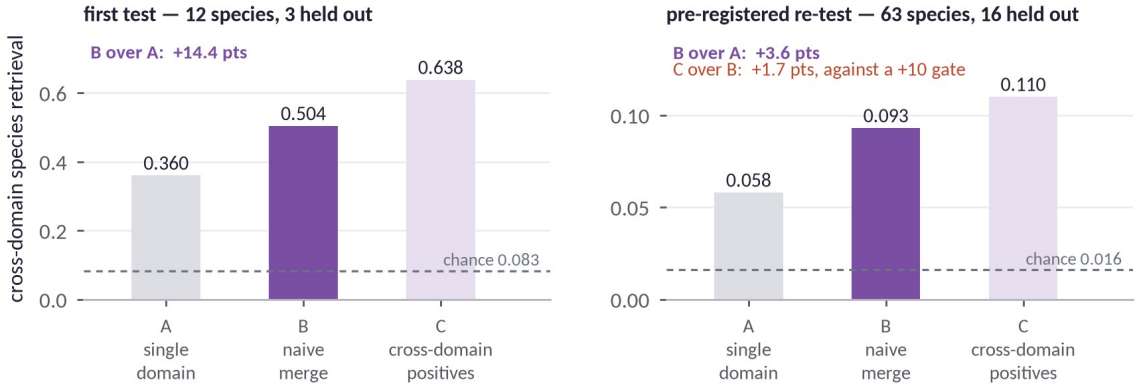


Figure 8 — The data road, and its honest shrink. Cross-domain species retrieval for three training arms, at two scales. Both are debug-scale, and the absolute levels are only comparable arm against arm inside each panel. Sources: docs/17 §17.4 and §17.6.

5 The map and the live path

> The map is a precomputed three-dimensional point cloud with one point per measured syllable. Where a measurement exists the map shows it. Where none exists, hand-set jitter fills the gap and carries the invented label. The app that reads the map is a static site. It loads JSON and float arrays, synthesises in an AudioWorklet, analyses the microphone, and runs entirely in the browser.

The browser runs the same oscillator as the pipeline and the same feature extractor on the microphone. The two renderers are verified sample for sample against each other at 18 of 18 parity fixtures, with the worst divergence at float32 rounding. Live audio is standardised, bridged into the map's space by eq. 6, and turned into cluster activations by a softmax over centroid cosines with about 120 ms of smoothing (docs/08 §8.2).

The map's honesty number is 0.61, and that number is a metric problem. Raw knot-level agreement between the bridged position and the true one is 0.61 on held-out recordings, which reads like a failing map. Decomposing the 38.6 % disagreement over 3 731 held-out syllables locates it. 76.4 % of it is boundary near-ties, with 74 % of all rows sitting within a margin of 0.05 and 41 % of flips landing on the runner-up knot. Tie-adjusted

agreement is 0.91. Big-margin map failure accounts for about 4 % of rows. At a knot boundary the question is ill-posed, so a better map class would leave the near-ties where they are. The single-label claim the interface makes is species-level agreement, which is 0.78 (docs/20 §20.1).

Eight recordings carry a quarter of the damage. Of 45 held-out sessions on the debug build, 8 carry 22.9 % of the map's flip loss, and all 8 are xeno-canto. They separate cleanly. An off-manifold 5-NN distance — computable from the 28-d side alone, which is all the browser will ever have — reaches AUC 0.953 against the bad-session label. A shipped cloud of 256 random anchors weighing 28 kB holds AUC at or above 0.92. The pre-registered two-signal combination reaches 0.954, so the single signal ships alone. The map-flip lever measured +3.5 points against a +5 bar, so the pre-registered fallback fired. The signal ships as an **honesty gate** rather than a correction: "this session is off everything the galaxy knows, and positions are guesses." On the shipped operating point it flags 8 of 45 sessions, catches 5 of the 8 bad ones including the three worst loss-carriers, and produces 3 clean false flags, all within 0.02 of the bad line (docs/20 §20.3.2).

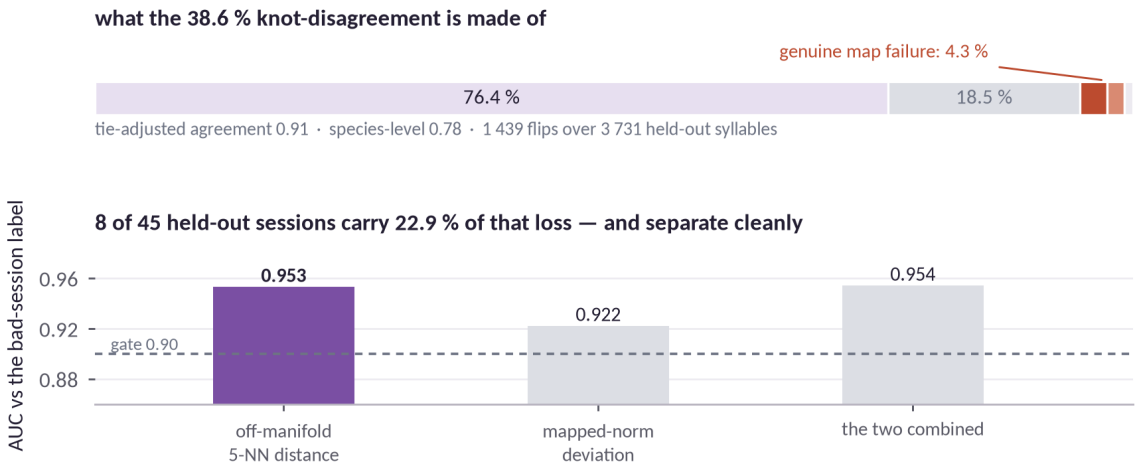


Figure 9 — The map, and the sessions that break it. Top: the 38.6 % knot disagreement decomposed, 1 439 flips over 3 731 held-out syllables. Bottom: AUC against the bad-session label for each candidate signal, with the pre-registered adoption gate. End-to-end debug build, debug-scale. Source: docs/20 §20.1 and §20.3.2.

Removing the bad sessions from the fit makes things worse. The obvious move is to drop the worst-quintile off-manifold sessions before fitting. It costs 0.006 and 0.015 of evaluation cosine on the two debug stores, against an allowance of 0.005, so it missed its gate. The reading is the useful part: the weird sessions are manifold coverage, and pruning them hurts the next unseen weird session. Gating belongs on the live path (docs/20 §20.3.1).

6 Structure above the syllable

> A syllable is not a sentence. Above the syllable sit transitions, bouts, song types and dialects, and this is where a project about birdsong either becomes interesting or admits it is a sound library. Birdy measures what it can at this level. Corpus depth is what starves most of it, and

> the machinery is already built and pre-registered. The re-test after the depth harvest is therefore a re-run.

The literature places birdsong above first order. Repeat-count adaptation and partially observable structure are both established[48,49]. Birdy ran the pre-registered bake-off anyway: the shipped first-order transitions against Jin–Kozhevnikov repeat-count structure and against a hidden-state HMM, scored on held-out bout likelihood. The repeat-count model lost decisively at −0.348 nats per token, winning 0 of 30 species. The HMM won the stated rule at +0.422 nats and 29 of 30 species. Then the decomposition reversed the adoption. On the 90 % of held-out transitions that were seen in training, first-order is the better model, −1.525 against −1.624. The HMM's entire advantage sits in the 10 % unseen tail, −9.25 against −4.16. That is a property of smoothing *shape*, and not of recovered hidden structure. First-order stays (docs/18 P1).

The finding was worth more than the bake-off. It named the real weakness as unseen-context transitions, at −9.25 nats per token, and that has a cheap fix. A Kneser–Ney continuation backoff gained +1.90 and +2.32 nats per token on unseen-context transitions across the two debug stores, while the seen side stayed bit-identical with loss of exactly 0.0000. The scope belongs next to it: 80 % of the unseen tail is seen-context, so the gain over the full tail is +0.37 and +0.65.

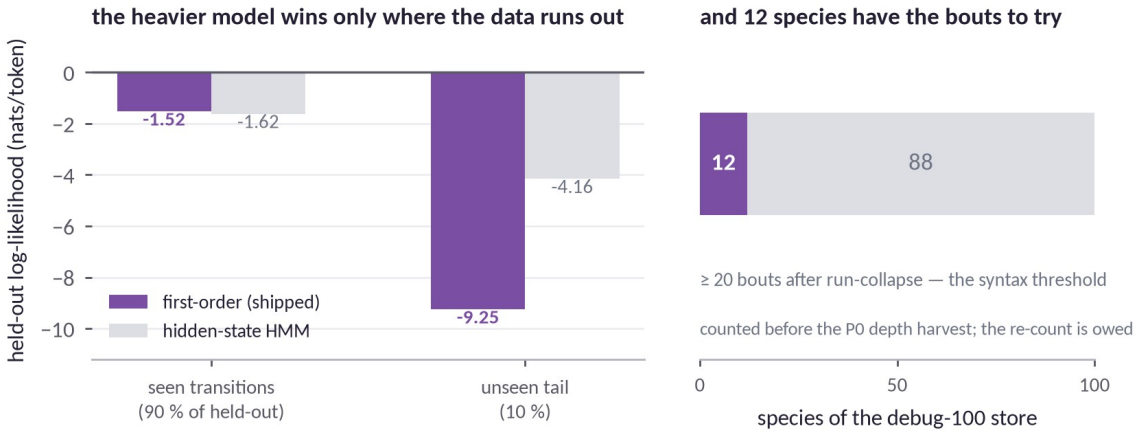


Figure 10 — Where the heavier grammar's advantage lives, and what syntax still needs. Left: held-out likelihood split by whether the transition was seen in training. Right: species of the debug-100 store with enough bouts to attempt song-level structure. Both debug-scale. Source: docs/18 P1 and P2.

Song types fall out of the data, with a control under them. Clustering run-collapsed sequences of type ids by normalised edit distance found 50 song types across 13 of 35 species, placing 693 of 2 419 candidate bouts. A global-permutation control yields 0.2 % at the shipped constants, which is one spurious type, so the recurrence floor is measured. The nightingale, the famous improviser, yields zero types over 275 candidate bouts, and that agrees with a century of field notes (docs/12). Forty-eight species carry detected song types today. The app plays each type heard at least twice as a song. Repeats are sampled from the bird's own measured self-transition rates, and the gaps come from the measured transition table.

The song-level kernel lost to a simpler baseline, and one number says why. A weighted edit distance over sequences of type and gap loses species lift to plain bag-of-types, +0.877 against +0.924 on the 37-species store and +0.050 against +0.081 on debug-100. It also misses the recordist gate at +0.264. Within-species song structure does exist, and all 18 deep species separate their own recordings. It lives entirely in the token marginals, so order and rhythm sit at the measurement floor at current depth. The one number is that only 12 of the debug-100 species have 20 or more bouts after run-collapse. The bar any learned song encoder must beat is now bag-of-types, and the depth harvest is what makes the attempt worth running (docs/18 P2).

7 The unknown bird

> What happens when a bird the system has never heard starts to sing? The arrival is measured and compared against everything known. A syllable that matches nothing is allowed to be new. It waits with similar arrivals until enough of them agree to found a new knot in the map. An external model may suggest which species it resembles, and that suggestion is metadata, never identity.

The arrival pipeline is live end to end. Each new observation is embedded and scored against a calibrated novelty threshold, which is set at a chosen false-positive rate. Known-looking syllables inherit the nearest type. In a relabel-recovery test, 97.4 % of relabelled syllables inherited a known type and 65.2 % inherited the correct species' type. The mis-assignments go to similar-sounding types of other species, which is the detector behaving as designed: it prices acoustic neighbourhood. Novel syllables wait. Below quorum they stay unassigned. Quorum is 8 or more coherent pending points inside a 0.70 cosine radius, above a 0.75 cohesion floor. At quorum they found a new knot labelled *unheard syllable*, at confidence **“unlabelled: real acoustics, no documented function”** (docs/10 §10.3 and §10.4, docs/16 §16.5).

A Perch 2.0 pass[43] labels the novel-flagged syllables from the training cache. It runs as ONNX, offline, at about 48 ms per window, and its label reads *closest documented species, Perch 2.0 inference, unverified*. The label is metadata, and the typing path never reads it. The store holds 23 697 novel syllables, and the pipeline reports 841 of them as unlabelable, because their xeno-canto audio was deleted by design. Structure above the syllable inherits the same label all the way up (docs/18 P6).

A threshold holds only for the corpus that calibrated it. A novelty threshold calibrated for 5 % false positives on the 9 k store flagged 29.8 % of the 47 k store. Recalibration is cheap and now runs automatically, and the ROC is reported per corpus epoch.

8 The ruler problem

> One number judges every fit decision in the pipeline: the inversion's gesture search, the respiration and two-voice ship gates, the fricative routing, the extension bake-offs. That number is the cosine between the resynthesis's 28-d descriptor and the real audio's. In 2026 the project measured the assumption under all of those gates. Is a cosine gain an audible gain? It is not. This section reports that measurement and its consequences, and it is the reason section 11 is ordered the way it is.

The bench (docs/19 §19.1) is the instrument. For a fitted type, render it, then take the *medoid* member's cached audio, which is a typical rendition rather than a lucky one. Resample the longer clip to the shorter, because time inside a syllable is relative. Shift the render's per-band log-mel mean onto the original, because channel is the measured dominant nuisance. Then score two independent spectral distances: **mel-cepstral distortion** over 25 coefficients in Kubichek dB, band-limited per pair to the file's own Nyquist, and **multi-resolution STFT error** at 512, 1024 and 2048. The self-test is the part that earns trust. Identity scores 0.000 dB. A half-low-passed render costs +77.9 dB MCD and +7.0 dB STFT while the cosine drops only 0.015.

The finding. Over 200 fitted types on the debug-100 store, the cosine's correlation with audible distortion sits at about 0.1. The p-values cannot reject zero, and the point estimates lean the wrong way. Two controls place the finding in the objective itself. The recomputed cosine reproduces the stored one at Pearson 0.921, so the bench measured the number the learn actually uses. MCD and STFT agree with each other at 0.925, so they measure something coherent, and the cosine tracks something else. There is even a plausible mechanism for the wrong-way lean. High-cosine types are clean whistles whose renders concentrate energy in one band while the field audio carries broadband noise everywhere else, so the spectral distance is large. Noisy buzzes score a low cosine and have broadband energy on both sides, so their distance is small.

The obvious fix was tried at two sample sizes, and the larger one settled it. A mel-path target — a time-resolved exp(−LSD) similarity to the type's measured mel trajectory — is the better *measurement* by a wide margin, at Spearman −0.420 against MCD where the cosine reads +0.100, signs correct. It is the worse *optimiser*. At blend weight 1.0 the cosine guardrail collapses by 0.377. A 10-type probe won 7 of 10 on MCD. The pre-registered 80-type gate run took that to a 31 % win share against a 60 % bar, with a median ΔMCD of exactly 0.0 dB and one type losing 0.120 cosine. The reading is better than the bare verdict. The objective change mostly leaves the fit where it was, because the inversion already lands where the blend would send it. It occasionally helps enormously, with a mean of +2.07 dB pulled by a handful of types at +13.8 dB. And it sometimes wanders. An objective with a flat median and an unguarded tail stays off by default (docs/19 §19.4 and §19.5).

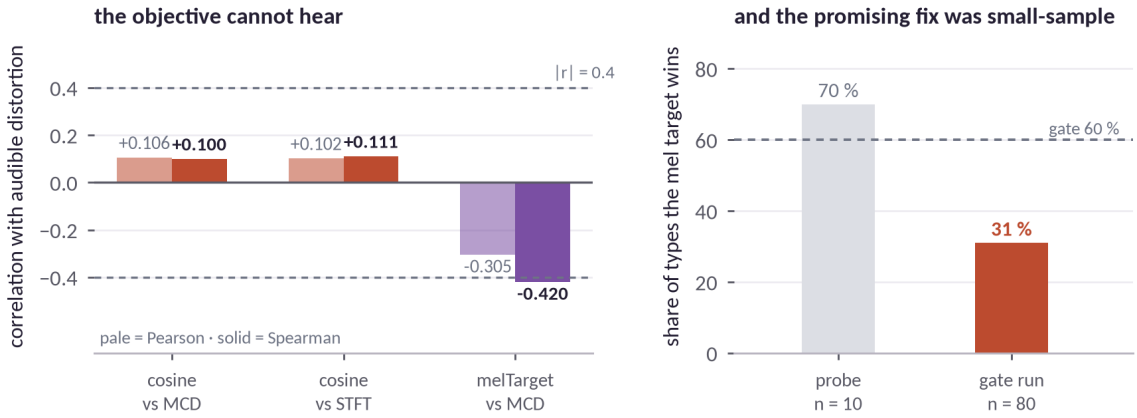


Figure 11 — The ruler problem, in two panels. Left: correlation with audible distortion for the incumbent objective and its replacement, n = 200 types. Right: the replacement's win share at the exploratory scale and at the pre-registered gate scale. Debug-scale, debug-100 store against the iNat cache. Source: docs/19 §19.2, §19.4 and §19.5.

The learned alternative failed earlier and harder. A realism-trained critic is an excellent classifier, at AUC 0.990 to 1.000 on entirely held-out species. Used as the fit objective it is monotonically harmful in its weight. At w = 1 the cosine collapses from 0.762 to 0.335 and placement drops to 17.3 %. The refutation replicated on the 40 k store, where placement went from 38.4 % to 31.9 % at w = 0.5. Perfect separation of two distributions and a gradient that points at the nearer one are different properties. The retained cosine guardrail is what made the gaming visible (docs/09 §9.7 and §9.8).

A structurally better tract tells the same story from a third angle. A fixed waveguide with 32 times the spectral freedom of the tracking band improved all three metrics at once on five species. Across 22 species it was a regression, at 23.1 % against 22.2 % placement. Later work traced that rejection to badly defined types, upstream of the objective. It now wins on the gan store under learned typing at 37.0 % and pays for it in cosine, so it stays promising and unconfirmed (docs/08 §8.1c, docs/09 §9.7).

The oldest open problem is species identity, and four candidate explanations are measured out. On the original 22-species store a resynthesis lands nearest its own species 23.1 % of the time against chance of 4.5 %. The envelopes keep their spread: the synthesiser expresses 103 % of the real between-species spread and 106 % of the spectral detail, varied along different axes. A spectral tilt error does exist, and correcting it exactly drops placement to 20.2 %. The harmonic structure is present, because real syllables resolve to a mean of 2.2 partials. And a time-resolved objective improves its own metric by 36 % while placement falls from 61.7 % to 43.2 %, which means the two objectives conflict. The ceiling sits in the descriptor and the filter chain's capacity, upstream of the search (docs/08 §8.1a).

The state of the metrology, verbatim from docs/19 §19.5: *“measure spectrally, optimise descriptor-ly”*. Every fit gate from here reads the bench. The inversion keeps climbing the cosine, because its failure is bounded and cluster-safe. An objective that is both audible and optimisable remains the open problem of this note.

9 The ledger

> One table, both directions. Every pre-registered gate this project has run, what it asked for before the run, what it measured, and what shipped. A gate written down in advance is the only kind whose result means anything, and a refutation that stays on the record is the only kind that stops somebody trying it again. Read the failures first. They are why the successes above can be taken at face value.

what was tried	the gate, set in advance	what it measured	verdict
Typing in the critic's 32-d embedding	placement up, cosine guardrail held	38.4 % → 51.7 %, cosine 0.807 → 0.796	**ships**
Parallel and streaming learn	bit-identical to serial	bit-identical, ~3 h → ~15 min	**ships**
Fricative route, noise through fitted formants	resynthesis beats the tonal fit, per type	bench 0.584 → 0.656, 13 wins / 0 losses	**ships per type**
Tracheal comb per species	comb improves the resynthesis	won 10 of 22 species	**ships where it wins**
Measured texture on authored one-tone calls	gain against the same bird's learned types	14 of 15, median +0.148	**ships**
Respiratory oscillator, hand-set default	beat the envelope on median cosine	0.343 → 0.336	**fails, parked**
Respiratory oscillator, per-species fit	win the type, gate held everywhere	won 445 of 12 680 scored types	**ships per species**
Two-voice coupling, hand-set default	beat the single voice	0.343 → 0.294	**fails, parked**
Two-voice coupling, per-species fit	win the type	won 2 027 of 5 922 screened	**ships per species**
RFF ridge browser bridge	median held-out cosine ≥ 0.8	0.962 on the debug build	**ships**
Linear ridge bridge, the incumbent	the same 0.8 gate	0.7997	**refuted, replaced**
Realism logit as the fit objective	placement and cosine hold	cosine 0.762 → 0.335, placement 17.3 %; replicated at 40 k	**refuted twice**
Mel-path objective at w = 0.5	≥ 60 % MCD wins, ΔMCD > +1 dB, cosine drop ≤ 0.02 / max 0.05	31 % wins, 0.0 dB, max drop 0.120	**fails, not adopted**
Mel-path as a measurement	$\|r\| \geq 0.4$ against MCD	Spearman −0.420, Pearson −0.305	**adopted as the bench target**
Channel-EQ + Freq-MixStyle augmentation	−30 % recordist lift, species held	−0.7 % recordist	**fails**
Recordist-adversarial head	the same	−33 % recordist for −14.7 pts species accuracy	**fails**
Within-species adversarial head	the same, species-safe by construction	species flat, recordist −6 %	**fails**
Naive two-domain merge	cross-domain retrieval up	+14.4 pts at 12 species, +3.6 at 63	**no adoption, direction holds**
Cross-domain contrastive positives	+10 pts over the merge	+13.4 at 12 species, +1.7 at 63	**fails on re-test**
Jin–Kozhevnikov repeat-count grammar	beat first-order on held-out bouts	−0.348 nats/token, 0 of 30 species	**refuted**
Hidden-state HMM grammar	the same	+0.422 nats overall, all of it in the 10 % unseen tail	**refuted on decomposition**
Kneser–Ney unseen-transition backoff	gain on unseen contexts, no loss on seen	+1.90 / +2.32 nats, seen side exactly 0.0000	**ships**

what was tried	the gate, set in advance	what it measured	verdict
Song-level edit-distance kernel	beat bag-of-types on species lift	+0.877 vs +0.924, and recordist lift +0.264	fails both gates
Song-type detection against a permutation null	recurrence above the null	0.2 % at the shipped constants, one spurious type	ships
Session gate: off-manifold 5-NN	AUC ≥ 0.90	0.953, and ≥ 0.92 from 256 shipped anchors	ships
Session gate: map-flip lever	+5 points of agreement	+3.5 points raw	fails, ships as an honesty gate
Session gate: prune bad sessions from the fit	evaluation cosine cost ≤ 0.005	−0.006 and −0.015	refuted
Learned inverse for synthesis parameters	2× faster at equal cosine	0.737 vs 0.734, and 0.2 s either way	fails its ship rule

Table 4 — The ledger. Twenty-eight pre-registered gates. Twelve ended in something shipping, and sixteen ended in a written-down refusal. Two of the refusals still produced an asset: the map-flip lever became an honesty gate, and the mel target became the bench instead of the objective. Sources: docs/08 to docs/21 as cited in the body.

The shape of that table is the argument. A project that reports only the twelve looks, from the outside, like a project that ran twelve experiments and got lucky. The sixteen are what make the twelve legible. Three of them cost real work and are worth naming as the expensive lessons: hand-set physics loses bake-offs, small samples flatter the arm you hoped for, and an objective can be excellent at telling two distributions apart while being useless at pointing toward the nearer one.

10 Where Birdy sits

> Birdy sits among bioacoustic tools that classify, embed, map or generate. Table 5 puts the approaches beside each other on the properties this project has to care about. A cell says “not documented” where the project's own references leave the question open.

approach	what it is	sings	interpretable parameters	ships in a static app	labels its claims
BirdNET	species classifier on spectrograms	no	no	no — CC BY-NC-SA, non-commercial	not documented
Perch 2.0	14 597-species embedding model	no	no	no — 409 MB ONNX, pipeline only	no
Google Bird Sounds	t-SNE map of thousands of recordings	no	no	yes — Apache 2.0 web experiment	no
Sainburg et al.	VAE latent space across species	no	latent, not physiological	not documented	no
Goffinet et al.	learned low-d features for individual and group differences	no	no	not documented	no
generative audio models	waveform generators	yes	no	no — training-data licences unclear	no
Birdy	physical syrinx + a corpus with no audio	yes	yes — pressure, tension, tract geometry	yes — no audio retained, no weights shipped	yes — five levels

Table 5 — Approaches beside each other. Sources for the external rows are references [42] to [45] and the project's docs/13. “Not documented” means the references do not answer the cell.

What Birdy takes, and what it declines. The map's prior art is open and was taken freely. A t-SNE map of bird recordings, a VAE latent space and learned low-dimensional feature spaces together established that vocal repertoires have navigable geometry[42,40,41]. Perch 2.0 gives novel syllables a second opinion, as metadata and never as identity[43]. BirdNET's non-commercial licence excludes it from the app[44]. A generative audio model would trade interpretability, the claim ladder and the playback-tested realism evidence for sample quality the field cannot yet measure. On the documented cells of table 5, Birdy stands alone in combining three properties: a bird-validated physical voice, a corpus that keeps no audio, and a label on every claim.

11 What comes next

> In order, each with the number that would prove it. The depth harvest that blocked most of this list is done, so the next steps are re-runs of pre-registered comparisons. Under all of them sits the metrology problem of section 8, which stays named and open.

Done: the depth harvest. 100 species, up to 30 recordings each, +139 440 syllables, median 29 recordings per species and 71 of 100 above the syntax threshold. That one action unblocked the grammar re-test, the song-level space and the widening of the spectral bench. It has the widest reach of anything on this list.

Next: the grammar bake-off, again. First-order against Jin–Kozhevnikov against a partially observable Markov model, on the deepened store, on the terms already written down. Thin data is exactly where higher-order models starve, and the repeat-count model's 0 of 30 says so. The bake-off is a re-run, and its result is pre-committed. **Then the song-level embedding.** Bouts as points. The parameter-free kernel baseline already lost, so the bar for any learned encoder is bag-of-types at +0.924, with the recordist-lift gate upstream of it (docs/18 P2).

Then structure, and only then anything resembling meaning. A CETI-style factorisation[50] would decompose each deep species' songs into independent axes — rhythm, tempo, ornamentation — toward per-species combinatorial alphabets, which is the strongest honest claim available without playback of Birdy's own. Dialect and individual maps follow: same-species variants placed geographically, individuals separated where 10 to 16 bouts each exist, which the literature has shown to be sufficient (docs/18 P3 and P4).

Two synthesis extensions, ranked and pre-registered. The audit of section 2.5 ranks the extension roadmap by realism per unit of interpretability, and the top two are already specified. First, the per-species respiratory pressure fit, forced by each type's measured gesture rather than a median envelope. Its gate is the bench: an MCD win share of 60 % or

better, a median ΔMCD better than 1 dB, a median cosine drop no worse than 0.02 and a maximum no worse than 0.05, measured on the debug-100 store against its cached audio. Those are the same gates that refused the mel-weight objective, applied deliberately. Second, measured physiological jitter and tremor per type: two scalars extending the existing roughness mechanism into a two-band measured gesture, against the roughly 1 % rendition-to-rendition frequency variation the literature reports[34,35]. Its gate is a KS distance halved against the unmodulated render, with no type regressing more than 1 dB of MCD. It runs on the 100 deepest species of the depth harvest. That is the first corpus with enough renditions to measure the target at all (docs/21 §21.3).

The road the project declines, and the reasons for it. Neural audio generation stays out: no DDSP decoder, no RAVE, no diffusion vocoder, no neural timbre transfer. Three reasons, in increasing order of weight. First, they give up parametric control and interpretable parameters. Second, they ship weights the browser contract forbids. Third, they cannot carry the claim ladder. Every claim in the interface needs a confidence label grounded in a mechanism a user can inspect, and a neural generator's mechanism is a weight matrix. The project's own evidence discipline makes the same argument from the other side: hand-set *physical* extensions, with five interpretable parameters and the literature's own equations, still had to lose two bake-offs in public before earning a fit pass. A generator that cannot state its loss in physical terms could never be judged that way. Learned components analyse and score. They never generate audio (docs/21 §21.4).

The open problem, named. Under the roadmap sits metrology. The project optimises one number, the descriptor cosine, which is cluster-safe and audibly blind. It measures with another, MCD and STFT, which are audibly faithful and hostile to an optimiser. It has refuted the learned third, the realism logit. Every future fit gate reads the bench, and every future objective candidate is pre-registered against it. Structure first, and meaning only from playback. That ordering is the roadmap's spine, and it does not move.

12 References

External literature, numbered in list order, resolving the superscripts in the body. Internal research notes are cited inline as “docs/19 §19.2”, which is how they are addressed in the repository. Of these, docs/08 is the normative contract that both halves of the codebase implement.

[1] Rutz, C. et al. “Using machine learning to decode animal communication”. Science, 2023. DOI 10.1126/science.adg7314.

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